Quiz 1

1. What is the need of minimum and maximum length restrictions for an Ethernet frame? Discuss what will happen if a station transmits 40 bytes of payload received from the upper layer?
2. Discuss how hidden and exposed terminal problems impact CSMA/CA performance. Further, discuss the techniques such as RTS/CTS handshaking, Network Allocation Vector (NAV), and backoff mechanisms to enhance overall throughput efficiency.

Quiz 2

1. For an IPv4 datagram, the value of header length is 5, the value of total length is 600, the more fragment bit (i.e., M) is set as 0, and the value of fragmentation offset is 200. Determine:
   d) The position of the datagram (first, last, middle, etc.)
   e) The sequence number of the first byte of the payload
   f) The sequence number of the last byte of the payload
2. Suppose five departments of an institute A, B, C, D and E need 110, 45, 28, 14 and 12 IP addresses. A class-C network address 198.131.172.0 has been allotted to the institute. Suggest the appropriate subnetting scheme and derive the appropriate subnet marks for each department. Also provide the starting and ending address of each department.

Quiz 4

1. In a network of 20 routers, each router generates ICMP Destination Unreachable messages for 0.5% of incoming packets. If 80,000 packets traverse the network per minute, calculate:
   a) Total ICMP Destination Unreachable messages generated.
   b) If only 70% of these messages reach the sender, compute the effective error feedback rate.
2. Differentiate between Distance Vector routing protocols and Link State routing protocols in terms of their operational mechanisms, convergence behaviour, routing information exchanged, susceptibility to routing loops, and scalability.

Quiz 5

1. A DHCP client receives an IP address with lease time = 24 hours. Using Forouzan's rules: Renewal time T1 = 0.5 × lease, and Rebinding time T2 = 0.875 × lease. Compute:
   a) T1 and T2 in hours and minutes
   b) If the client rebooted at the 12th hour, calculate how many hours it can still use the assigned IP before rebinding.
2. Given the steps involved in DNS resolution (recursive and iterative queries), trace the full DNS lookup process for the domain: cse.manit.ac.in. Assume the local DNS cache is empty. Identify each DNS server involved and illustrate the query–response path.

Name of the Student..... **Scholar Number**.....2521201110

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: Oct-Nov 2025**

Course: M.Tech **Semester:** I **Subject Code:** ~~CS 324~~ CS CN 1102
Subject Name: TCP/IP **Maximum Marks:** 20
Duration: 60 mins **Date:** 04/11/2025 **Time:** 9:30 AM to 10:30 AM

Note: Attempt all questions. Assume missing data (if any) and mention the assumptions.

Q. No.	Questions	Marks	COs
1.	a) Explain the significance and operational role of the Network Allocation Vector (NAV) in the CSMA/CA protocol. Further, analyze the impact and possible network behavior when RTS (Request to Send) or CTS (Clear to Send) control frames experience a collision during transmission. b) Provide a detailed explanation of the design rationale behind the restriction that prevents the generation of an ICMP message in response to another ICMP error message. Discuss how this rule helps maintain network stability and prevents potential issues.	3+2=5	CO1
2.	An organization has been assigned the IP address block 192.168.50.0/24. The network administrator must design a subnetting plan using Variable Length Subnet Masks (VLSM) to meet the following departmental requirements: Department: Research, Required Hosts: 60, Network Name: Net-A Department: Engineering, Required Hosts: 28, Network Name: Net-B Department: Sales, Required Hosts: 12, Network Name: Net-C Department: HR, Required Hosts: 6, Network Name: Net-D Department: Admin, Required Hosts: 2, Network Name: Net-E a) Allocate appropriate subnet addresses to each department using VLSM so that address space wastage is minimized. b) For each subnet, determine: i) Subnet mask (in both slash and dotted-decimal form) ii) Network address iii) First usable IP address iv) Last usable IP address v) Broadcast address	5	CO2
3.	An IPv4 router receives an IP datagram that carries 2000 bytes of data. The header length (HLEN) field in the IP header has a value of 5, and the router must forward this datagram over a link with a Maximum Transmission Unit (MTU) of 150 bytes. Assuming standard IPv4 fragmentation rules, determine: a) The number of fragments the datagram will be divided into. b) The data per fragment (for all fragments except the last). c) The total bytes sent (including all headers) after fragmentation.	5	CO2
4.	Explain in detail the working mechanism of the DHCP IP address allocation process from the moment a host connects to a network until it successfully obtains an IP address. Illustrate your answer with a suitable diagram showing the message exchange sequence between the client and server. Also, describe what happens if multiple DHCP servers respond to a client's request simultaneously?	5	CO3

**********END**********

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: END TERM **MONTH and YEAR: December 2025**

Course: M.Tech (CN) **Semester:** I **Subject Code:** CS CN 1102
Subject Name: TCP/IP **Maximum Marks:** 40
Duration: 120 mins **Date:** 12/12/2025 **Time:** 02:30 PM to 04:30 PM

Note: Attempt all questions. Assume missing data (if any) and mention the assumptions.

Q. No	Questions	Marks	COs
1.	(i) Why is the CSMA/CD protocol not suitable for wireless LANs, and why does IEEE 802.11 use CSMA/CA instead? (ii) With appropriate diagrams, explain the hidden-station and exposed-station problems in wireless LANs operating under CSMA/CA. (iii) Clearly explain how hidden and exposed station problems are addressed in wireless LANs.	2+4+4= 10	CO1
2.	Consider an IPv4 packet with option field(s) of 20 bytes. Calculate the checksum field for the IPv4 header when *service type* field is 0, *total length* field is 60, *identification* field is 1, *do not fragment* (*D*) bits is set to 1, *TTL* is 4, the higher layer protocol is TCP, initial value of the checksum is set to 0 at the sender, the *source address* is 5.11.7.5 and the *destination address* is 10.10.8.4.	5	CO2
3.	An ICMP message has arrived with the header (in hexadecimal): **05 00 11 12 11 0B 03 02** What is the type of the message? What is the code? What is the purpose of the message? What is the value of the last 4 bytes? What do the last bytes signify?	5	CO2
4.	For reliable frame transmission, a sender uses Stop-and-Wait ARQ protocol. Frames are 500 bytes in size, and the sender's transmission rate is 40 Kbps (1Kbps = 1000 bits per second). The size of an acknowledgement is 50 bytes, and transmission rate of the receiver is 2 Kbps. The propagation delay of the channel is 100 milliseconds. Calculate the throughput assuming no frames are lost.	5	CO2
5.	Provide a diagram illustrating TCP connection establishment using the 3-way handshake. Assume the initial sequence numbers chosen by the client and server are 1000 and 5000, respectively. Show all three segments exchanged during the connection establishment and clearly label the sequence numbers, acknowledgment numbers, and control flags in each segment. The segments exchanged are as follows: • From client to server: Seq = 1000, SYN = 1 • From server to client: Seq = 5000, ack = 1001, ACK = SYN • From client to server: Seq = 1001, ack = 5001, ACK = 1	5	CO2
6.	Describe the format of the SCTP common header and explain the function of each field it contains. Using a suitable diagram, illustrate how multiple SCTP chunks are encapsulated within a single SCTP packet. Clearly describe the roles of the Source Port, Destination Port, Verification Tag, and Checksum fields in ensuring reliable and secure communication.	5	CO3
7.	Compare and contrast the IPv4 and the IPv6 header fields. Do they have any fields in common? Explain different techniques for migrating from IPv4 to IPv6.	5	CO3

**********END**********

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL – 462003

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

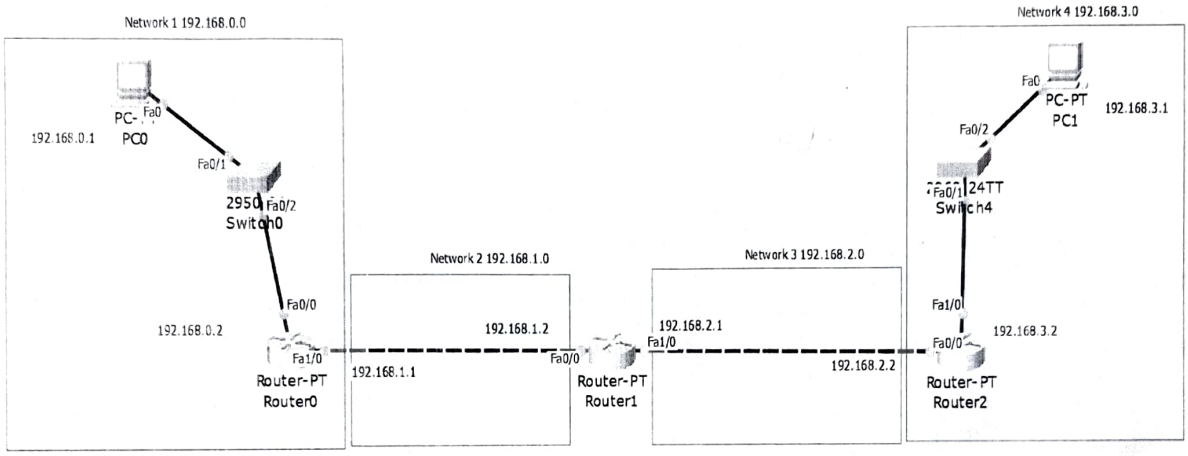
End Term Practical Examination, December – 2025

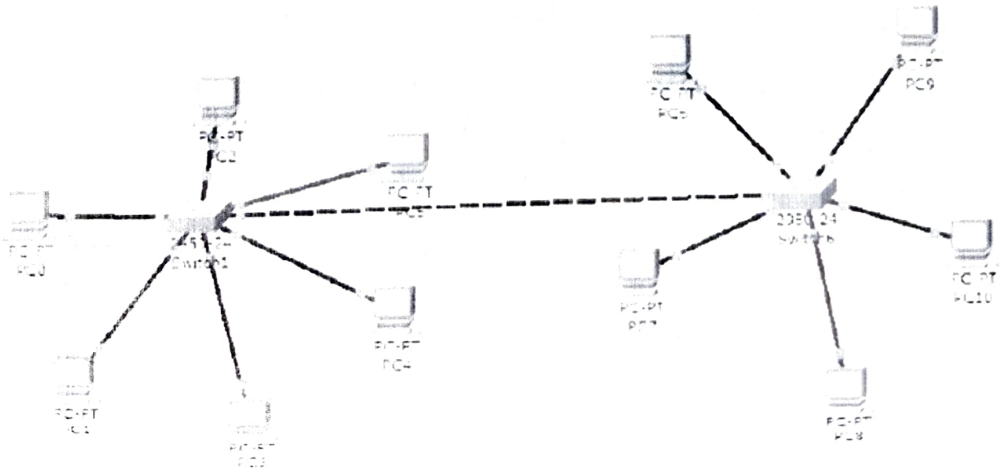
Core Subject Lab (TCP/IP) (CS CN 1104)
M. Tech. – I Semester

Date: 01/12/2025 Time: 11:30 AM - 12:30 PM

[Max Marks 40]

Objective 1: To demonstrate dynamic routing using the Routing Information Protocol (RIP) in a Cisco packet tracer.



The total number of networks are four: Network 1 is 192.168.0.0, Network 2 is 192.168.1.0, Network 3 is 192.168.2.0, Network 4 is 192.168.3.0

Objective 2: Configure the network for STAR topology using Cisco packet trace.